The Mitochondrial Calcium Uniporter Couples Hippocampal Burst Firing with Homeostatic Plasticity

1 Introduction

Mitochondria participate in diverse physiological processes through their handling of intracellular Ca2+ (*1–3*). The primary route of mitochondrial Ca^{2+} (mitoCa^{2+}) influx is the mitochondrial calcium uniporter (MCU) (*4*, *5*). MCU is indispensable for mitoCa^{2+} influx under evoked, high stimulation conditions (*4–26*). Importantly, there are no compensatory alternatives for evoked mitoCa^{2+} uptake, which is equally impaired by acute and chronic perturbations to MCU. Notwithstanding, MCU is highly conserved across all animals (*27*, *28*), while MCU homologs have been identified in nearly all plants and metazoa (*29–31*). Given its singular function and ubiquitous expression, one of the most surprising findings of the molecular era (*30*) is that MCU knockout (MCU-KO) typically results in remarkably mild phenotypes. Most organisms or tissues lacking MCU exhibit no gross pathology or dysfunction (*6*, *10*, *12*, *32–34*), which poses an 'evolutionary paradox' (*30*) – Why is MCU deeply conserved if its absence is well-tolerated?

A potential answer emerges when perturbing MCU and subsequently challenging the system beyond its resting state. Under these circumstances, the functional impairments that occur collectively suggest a fundamental role of MCU in adaptation; enabling cells, tissues, and organisms to effectively cope with various forms of physiological stress (*10*, *12*, *15*, *31–37*). MCU-KO in *Caenorhabditis elegans* leads to impaired epidermal wound healing (*32*), in *zebrafish* to greater incidence of sinoatrial arrest (*33*), in *Drosophila* to shortened lifespan (*10*), in *Arabidopsis thaliana* to reduced root length under low-nutrient conditions (*31*), and in mice to a diminished capacity for sustained physical exercise (*12*, *34*). This pattern is further solidified through murine cell-type conditional MCU-KO, which in the heart leads to blunted cardiac output in response to adrenergic stimulation (*35*), in skeletal muscle to diminished sprint performance (*37*), in pancreatic beta-cells to impaired glucose-stimulated insulin secretion (*15*), in the liver to exacerbated steatosis under high-fat diet conditions (*36*) and in the brain to impairments in short- and long-term synaptic plasticity (*11*, *17–20*, *22*, *24*, *38–43*).

The brain has adaptive needs beyond synaptic plasticity. To balance the continuous barrage of stimuli and plastic events without succumbing to hyper- or hypo-activity, neuronal networks rely heavily on homeostatic regulation (*44*). Homeostasis refers to the processes that stabilize key physiological variables within a healthy and functional range (*45*). Firing rates are homeostatically regulated (*46–51*), and the clinical importance of firing rate homeostasis (FRH) is increasingly evident (*51–54*). Still, the molecular sensors that track deviations in firing remain elusive. One possible candidate is MCU (*47*, *55*). Due to its activation kinetics, MCU preferentially responds to high-frequency burst firing (*41*, *47*, *56–58*). Neuronal bursts are a privileged mode of communication, essential for reliable synaptic transmission, the induction of synaptic plasticity, and the encoding of memories (*59*). The selectivity of MCU for burst firing and its position within the interconnected mitochondrial network allows mitoCa2+ influx to integrate salient activity across time and space, offering a robust representation of recent cellular history.

To test the functional relationships between MCU, burst-firing, and FRH, we performed a multi-level investigation of conditional MCU-KO in the mouse hippocampus, including long-term electrophysiological recordings alongside continuous pharmacological perturbations. Ultimately, we offer a new perspective on the evolutionary paradox of MCU by establishing its role in converting high-intensity signals into long-lasting adaptive responses, in particular homeostatic plasticity.

2 Results

We generated mice with targeted deletion of MCU in excitatory CA1 neurons and confirmed reduced MCU mRNA expression (Figure 1**Error! Reference source not found.**A,B). As expected, Ca^{2+} imaging in cultured hippocampal neurons revealed a deficit in mitochondrial Ca^{2+} uptake of MCU-KO neurons, evoked by high frequency trains of action potentials (Figure 1C). Evoked cytosolic Ca^{2+} transients remained indistinguishable from control (Figure 1**Error! Reference source not found.**D). We then shifted our focus to calcium dynamics during spontaneous neuronal activity (Figure 1E,I). Surprisingly, the Ca2+ transients of MCU-KO neurons were fully preserved in the mitochondrial compartment (Figure 1F-H), yet larger and more frequent in the cytosolic compartment (Figure 1J-L). Thus, while MCU is strictly required for rapid calcium uptake during intense stimulation, it can be compensated for during resting states by alternative, possibly slower pathways (*8*, *60–63*). The compensation, however, involves enhanced cytosolic drive.

To determine if the compensatory cytosolic drive in MCU-KO stems from a reconfiguration of baseline firing, we performed MEA recordings in cultured hippocampal neurons. While overall firing rates remained indistinguishable from controls, MCU-KO neurons exhibited significantly higher burstiness, defined as the probability of spikes occurring within bursts (Figure 2A). The same functional reorganization was observed in vivo. Extracellular tetrode recordings of regular spiking, putative pyramidal cells in CA1 of freely moving mice (Supp. Figure 1) revealed a shift from tonic towards phasic firing in MCU-KO neurons, with preservation of overall firing rates (Figure 2B).

Given burst firing of pyramidal cells is strongly associated with sharp-wave ripples (SWRs; (*64*)), we hypothesized that the elevated intrinsic burstiness of MCU-KO neurons would distort these network oscillations. Indeed, SWR in MCU-KO mice had an increased peak amplitude (Figure 2C), possessed a faster internal frequency (Figure 2D), and were significantly shorter in duration (Figure 2E). Furthermore, per-unit analysis of firing during SWRs revealed advanced timing of spikes in MCU-KO neurons; their spike center of mass relative to the SWR peak shifted towards earlier, negative time lags compared to control (Figure 2F,G). To determine whether this advanced recruitment is directly linked to the transition towards phasic firing, we modeled the spike center of mass as a function of baseline burstiness and firing rate. Earlier spiking during ripples was positively correlated with burstiness (Figure 2G), while firing rate had no significant effect (Supp. Figure 2). This relationship was conserved across genotypes, as interactions were tested and found non-significant (data not shown). These observations suggest that the loss of MCU triggers a shift toward phasic firing that distorts the temporal dynamics of synchronized network events, particularly SWRs (*65*).

Next, we examined the role of MCU in the canonical CA3-CA1 Schaffer collateral pathway using field excitatory postsynaptic potential (fEPSP) recordings in acute hippocampal slices (Figure 3A). Basal synaptic transmission, measured as the fEPSP slope relative to the presynaptic fiber volley amplitude, was enhanced in MCU-KO slices (Figure 3B). In addition, MCU-KO slices exhibited reduced short-term synaptic plasticity, evident by a significantly lower paired-pulse ratio (Figure 3C), and impaired long-term potentiation following theta-burst stimulation (Figure 3D). These findings indicate an increase in release probability (*66*) and total excitatory drive to CA1, which likely contributes to the enhanced baseline burstiness and cytosolic Ca2+ transients observed in MCU-KO neurons.

Having established the baseline phenotype of MCU deletion in CA1, we next asked how this altered network copes with prolonged functional stress. To test the capacity for FRH in the intact brain, we continuously delivered the $GABA_B$ receptor agonist baclofen via osmotic pumps directed at CA1 (Figure 3E). While baclofen suppressed the firing rates of both control and MCU-KO neurons initially, their recovery trajectories gradually diverged (Figure 3F).

Control mice successfully engaged homeostatic mechanisms, recovering steady-state firing rates back to baseline levels by the third day of the perturbation (Figure 3G). In contrast, MCU-KO firing rates remained significantly depressed throughout the entire perturbation (Figure 3G).

While all MCU-KO animals failed to recover their firing rates, we observed significant heterogeneity between individual animals and hypothesized that the initial state of the network predicts its capacity for FRH. Baseline mean firing rate, averaged across units per animal, did not predict recovery for either genotype (Figure 3H). In contrast, high average burstiness at baseline predicted robust recovery in control, but not MCU-KO mice (Figure 3I). These findings imply that at the network level, a preference towards phasic firing is associated with homeostatic resilience in a manner dependent on MCU.

Interestingly, while control neurons exhibited equal burstiness at baseline and steady-state, the homeostatic challenge with baclofen altered the firing pattern of MCU-KO neurons and collapsed their compensatory state of elevated burstiness (Figure 3J). However, these *in vivo* findings reflect population distributions since individual unit identities could not be maintained throughout the recording period. To investigate if burstiness also carries prognostic meaning for homeostatic recovery at the single-cell level, and better understand why this relationship fails in MCU-KO, we replicated the homeostatic experiment *in vitro* using MEA recordings, which allow tracking the same units over days with a high level of certainty.

Over the course of two days with baclofen, MCU-KO cultures exhibited significantly impaired FRH compared to controls (Supp. Figure X), confirming a deficit in homeostatic plasticity that is intrinsic to the hippocampal network. This complements our previous work showing that overexpressing the MCU complex rescues both burst-evoked mitoCa2+ transients and FRH in neurons lacking the Insulin-like Growth Factor-1 Receptor (IGF-1R), an upstream promoter of MCU (*47*). Paired with the genetic ablation presented here, these results forge a definitive causal link between MCU-mediated mitoCa2+ signaling and the homeostatic response to baclofen.

Next, we dissected per-unit firing rates into two components: single-spike firing and burst-spike firing. During the baclofen perturbation, both firing modes followed similar trajectories over time within each genotypes (Figure 4A). Single-spike and burst-spike firing gradually recovered in controls but were persistently suppressed in MCU-KO networks. This parallel behavior, where the two components rise and fall together, prompted us to ask whether homeostatic mechanisms merely scale activity irrespective of its pattern, or if there exists a coordinated exchange rate between tonic and phasic firing.

To visualize how these components are co-modulated during FRH, we plotted the absolute difference from steady-state to baseline in single-spike firing versus burst-spike firing and fitted the data using orthogonal regression (Figure 4B). In Control neurons, variance was distributed across both axes, demonstrating a tightly coupled modulation where absolute changes in burst firing are accompanied by similar absolute changes in single-spike firing. In contrast, the variance in firing rate recovery of MCU-KO neurons was almost entirely restricted to the burst-firing axis. This apparent breakdown in coordination raised a critical question: does the absence of MCU sever the coupling between these two firing modes, or does the coupling exist but fail to successfully execute?

To answer this question, we transitioned from the absolute to the relative domain. First, we normalized each unit to its own pre-perturbation baseline, and examined the temporal evolution of single-spike and burst-spike firing throughout the perturbation (Figure 4C). This relative perspective revealed temporal dynamics that were markedly different from the parallel trajectories initially observed in the raw population averages (Figure 4A). Rather than simply rising and falling in tandem, the normalized trajectories exposed a strategic divergence during

the homeostatic response. Burst-spike firing of control neurons overshot baseline levels, presumably to compensate for a recovering, yet slightly suppressed, single-spike rate.

To formalize this apparent relationship, we analyzed the state-space trajectories of single- and burst-spike firing gain, where gain is calculated as the log-fold ratio between steady-state and baseline (Figure 4D). Remarkably, both genotypes obeyed a conserved, statistically identical, power-law scaling. For every 10% change in burst-firing during the perturbation, control and MCU-KO neurons exhibited a 4.3% and 3.9% change in single-spike firing, respectively. To ensure that the power-law scaling between burst and single-spike recovery is not a byproduct of differences in baseline activity, we implemented a partial regression analysis. Even after regressing out baseline firing rate and burstiness, the proportional allocation of spikes during FRH persisted (Supp. Figure 3C). Moreover, evaluating the log-fold gain of aggregate mean rates per animal *in vivo* revealed a similar positive scaling relationship between burst-spike and single-spike recovery (Supp. Figure 3D). Together, these results imply that baclofen-induced firing rate homeostasis involves coordinated plasticity of tonic and phasic firing, both *in vitro* at the single-cell level and *in vivo* at the network level. Importantly, MCU-KO neurons follow the same power-low governing the exchange of burst- and single-spikes as control neurons, but they fail to execute it.

The discovery of a conserved power-law scaling in the relative domain resolves the apparent decoupling observed in absolute firing rates. Because the allocation of burst- and single-spikes operates proportionally, the divergence between genotypes in the absolute domain is a mathematical consequence of the burst-dominant baseline in MCU-KO neurons relative to control neurons. Applying any proportional modulation to this baseline during a homeostatic challenge will naturally force a greater absolute change in burst spikes compared to single spikes (Supp. Note 1). However, while this geometric relationship explains how the spike deficit is allocated, it does not explain the deficit itself. The conserved power-law dictates how burst and single spikes scale together, yet it does not guarantee that the network will generate enough overall activity to restore homeostasis. Thus, we shifted our focus from the plastic adaptation of firing patterns during FRH, to the biological capacity for executing homeostasis itself. Specifically, we hypothesized that the homeostatic deficit caused by MCU deletion is directly linked to the compensatory increase in phasic firing.

To visualize how the initial network state influences a neuron's capacity for recovery, we stratified units into subgroups based on their baseline burst probability and tracked their normalized firing dynamics throughout the homeostatic challenge (Figure 4E). In Control networks, a clear prognostic hierarchy emerged: neurons entering the perturbation with a higher baseline burstiness mounted the most rapid and robust homeostatic rebound. Conversely, control neurons with low baseline burstiness exhibited a much more sluggish and incomplete recovery. This prognostic hierarchy appeared completely inverted in MCU-KO networks, where neurons exhibiting the highest initial burstiness suffered the most profound firing rate depression.

To statistically quantify this relationship and disentangle the volume of activity from its temporal pattern, we modeled steady-state firing as a function of baseline firing rate and burst probability, extracting the marginal effect for each metric. Baseline firing rate served as a consistent positive predictor of absolute steady-state activity, exhibiting no significant interaction with genotype (Supp. Figure 4A). Neurons that were more active initially simply remained more active after the perturbation. In stark contrast, the marginal effect of baseline burstiness exhibited profound interactions with genotype (Supp. Figure 4B). In Control networks, higher baseline burstiness independently predicted robust recovery of single-spike, burst-spike, and total firing rates. In MCU-KO networks, these prognostic relationships were significantly impaired, and even inverted for single-spike recovery. MCU-KO neurons with the highest initial burst probability exhibited the most severe collapse in tonic firing.

While marginal effects successfully isolate the prognostic relationship of baseline burstiness from the that of firing rate, they treat the recovery of single and burst spikes as independent outcomes. However, our state-space analysis demonstrated that these two firing modes are coupled by a conserved allocation power-law. This interdependence raises a critical mechanistic question: does the compensatory burstiness of MCU-KO neurons directly suppress their capacity to recover tonic firing, or does the baseline state primarily suppress the recovery of phasic firing, which subsequently drags down tonic firing through their obligatory coupling? To formally distinguish between a direct suppressive effect of the baseline state and an indirect effect mediated by the allocation power-law, we performed a causal mediation analysis evaluating the relationship between baseline burstiness, steady-state single-spike firing, and steady-state burst-spike firing (Figure 4F and Supp. Figure 4C-F). In both genotypes, baseline burstiness exhibited a negative direct effect on single-spike firing and a positive indirect effect mediated by burst-spike firing. However, the negative direct effect was twice as strong in MCU-KO neurons, effectively outweighing the positive indirect effect. Consequently, the net total effect of baseline burstiness shifted from positive in controls to negative in MCU-KO neurons, resulting in a failure to recover tonic firing levels.

To definitively quantify the prognostic value of burst- versus single-spike firing, we performed a feature ablation analysis using k-fold cross-validation, evaluating the unique out-of-sample variance in steady-state firing explained by each baseline firing component (Figure 4G). Consistent with the privileged role of bursting in driving network plasticity, baseline burst-spike firing contributed significantly more unique prognostic information than single-spike firing across both genotypes. However, the predictive power of burst-spike firing was significantly degraded in MCU-KO networks compared to Controls. Overall, these findings establish phasic firing as a primary feature dictating a neuron's homeostatic resilience. However, they also reveal that the compensatory shift toward extreme burstiness adopted by MCU-KO networks is profoundly maladaptive; rather than promoting resilience, this altered baseline state suppresses the recovery of tonic firing and compromises the adaptive information carried by phasic firing.

To ensure that these physiological conclusions were independent of a specific analytical approach, we confirmed our findings hold across a wide range of burst detection parameters (Supp. Figure 5). A comprehensive parameter sweep of varying inter-spike interval (ISI) thresholds and minimum spike counts demonstrated that the unique prognostic value of baseline burst-spike firing remains remarkably stable regardless of the specific criteria used to define a burst. Furthermore, replicating our in vivo analyses using an autocorrelogram-derived Burst Index or applying the MEA dataset criteria yielded the exact same homeostatic allocation rules and prognostic relationships.

Our physiological findings present a complex picture regarding the impact of MCU deletion. On the one hand, MCU-KO exhibited profound impairments in synaptic and homeostatic plasticity, as well as significantly shorter SWRs – a feature directly linked to working memory impairments (*65*). On the other hand, baseline firing rates remained remarkably stable, supported by a highly burst-dominant state. To resolve whether this altered baseline state was sufficient to support complex behaviors, or if the underlying loss of plasticity and long-duration SWRs would prove dominant, we subjected MCU-KO mice and their littermate controls to a battery of hippocampus-dependent memory tasks. To ensure a clear interpretation of these cognitive assays, we first confirmed that the mice did not exhibit baseline differences in locomotor activity or anxiety-like behaviors that could otherwise confound performance (Supp. Figure 6).

In the novel object recognition task, control mice showed a robust preference for exploring a novel object over a familiar one, whereas MCU-KO mice explored both objects equally, indicating a failure to form or retrieve the memory of the familiar object (Figure 5A,B).

A similar impairment emerged in a social recognition paradigm, where MCU-KO mice failed to preferentially investigate a novel mouse over a familiar one (Figure 5C,D). These memory deficits also extended to associative learning. In a contextual fear conditioning task, MCU-KO mice displayed significantly less freezing behavior when re-exposed to an arena associated with an electric shock, demonstrating impaired memory of the fearful environment (Figure 5E-G). Crucially, these mnemonic impairments were specific to tasks reliant on the hippocampus. When tested in an auditory cued fear conditioning paradigm – where memory for a tone does not depend on hippocampal integrity – MCU-KO mice showed freezing levels that were indistinguishable from controls (Figure 5H-J). Together, these cognitive deficits demonstrate that MCU-mediated Ca^{2+} signaling is indispensable for the neural computations involved in hippocampal-dependent memory.

3 Figures

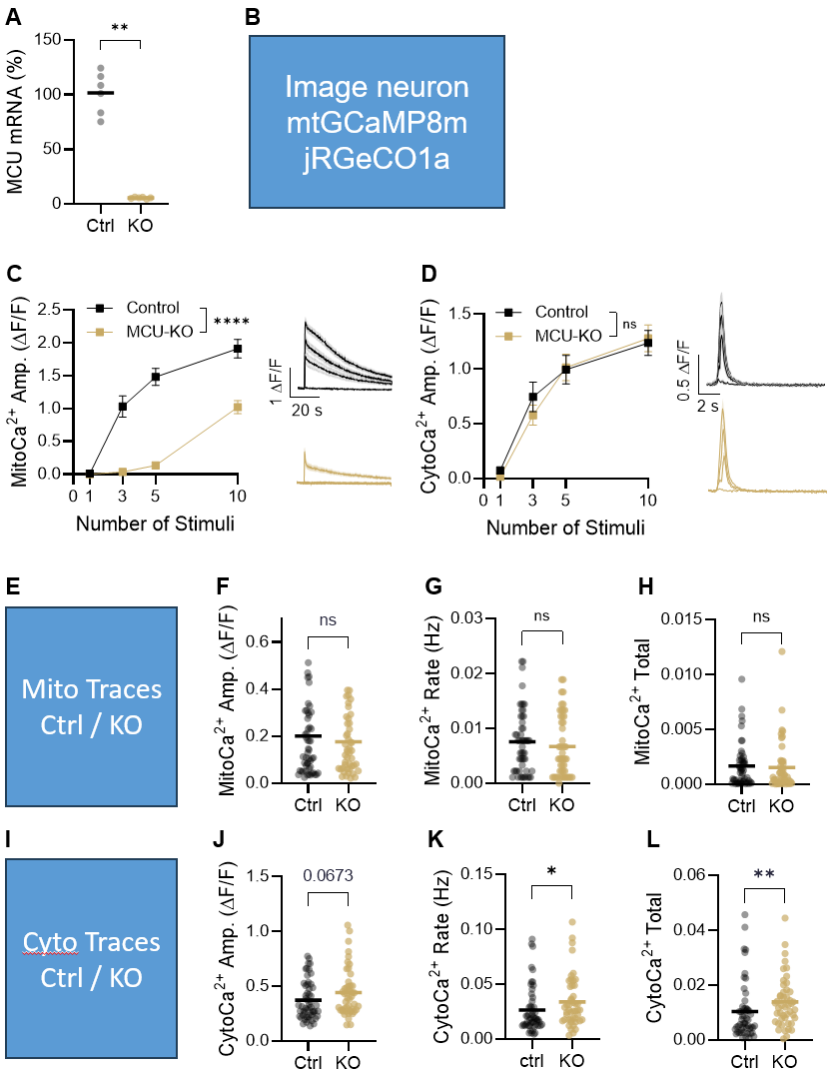


Figure 1. Elevated cytosolic Ca2+ transients compensates for MCU-mediated influx during baseline activity.

(A) Quantification of MCU mRNA expression levels in the hippocampus of Control and MCU-KO mice, expressed as a percentage relative to control levels. Individual data points represent mice (n = 6). The genetic knockout resulted in significantly lower MCU mRNA expression (p = 0.002).

(B) Representative image of a neuron expressing mtGCaMP8m-jRGeCO1a

(C) Peak mitochondrial Ca2+ responses (mitoCa2+, ΔF/F) in Control (n = 36) and MCU-KO (n = 29) neurons evoked by trains of 1, 3, 5, and 10 stimuli at 50 Hz. Mitochondrial Ca2+ uptake was significantly reduced in MCU-KO neurons compared to Controls at 3, 5, and 10 stimuli (p < 0.001). Right panel shows representative traces of mitoCa2+ responses in a Control (top) and MCU-KO (bottom) neuron.

(D) Peak cytosolic Ca2+ responses (cytoCa2+, ΔF/F) in Control (n = 32) and MCU-KO (n = 45) neurons evoked by trains of 1, 3, 5, and 10 stimuli at 50 Hz. No significant difference between genotypes was observed (p = 0.78). Right panel shows representative traces of cytoCa2+ responses in a Control (top) and MCU-KO (bottom) neuron.

(E) Representative traces of spontaneous mitochondrial Ca2+ transients in Control and MCU-KO neurons.

(F) Peak amplitude (ΔF/F) of spontaneous mitochondrial Ca2+ transients was not significantly different between Control (n = NN) and MCU-KO (n = NN) neurons (p = PP).

(G) Average rate (Hz) of spontaneous mitochondrial Ca2+ transients was not significantly altered by MCU deletion (p = PP).

(H) Total mitochondrial Ca2+ flux, calculated as the integrated area under the curve for spontaneous events, was not altered by MCU deletion (p = PP).

(I) Representative traces of spontaneous cytosolic Ca2+ transients in Control and MCU-KO neurons.

(J) Peak amplitude (ΔF/F) of spontaneous cytosolic Ca2+ transients was significantly elevated in MCU-KO (n = NN) relative to Control (n = NN) neurons (p = PP).

(K) Average rate (Hz) of spontaneous cytosolic Ca2+ transients was significantly elevated in the absence of MCU (p = PP).

(L) Total cytosolic Ca2+ accumulation was significantly elevated in the absence of MCU (p = 0.005).

Panel (A, F-H, J-L) analyzed using the Mann-Whitney test. Panels (C,D) analyzed using two-way RM ANOVA with Sidak's multiple comparisons test.

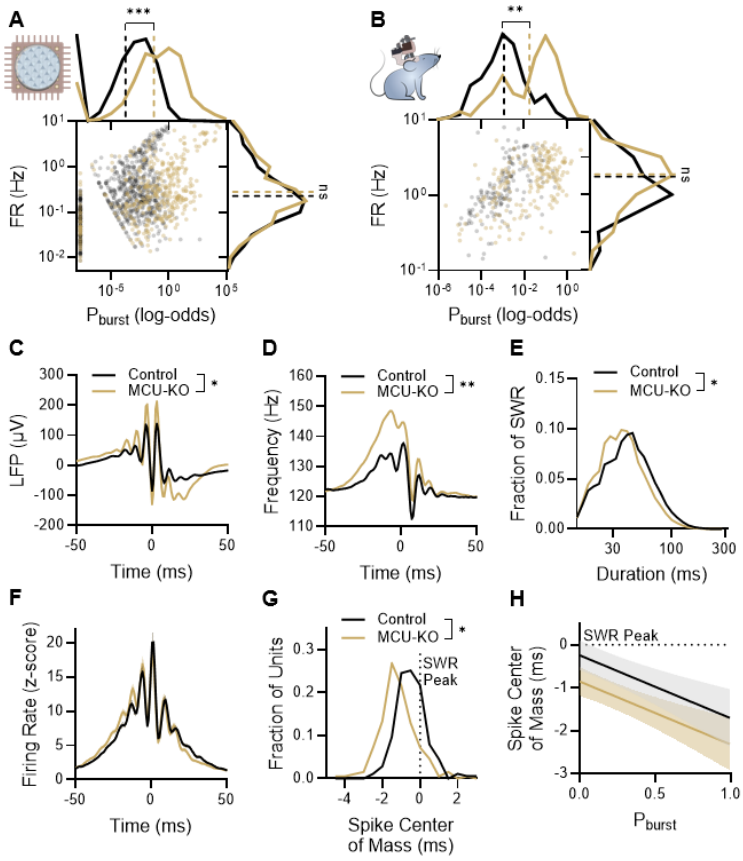


Figure 2. MCU-KO increases burstiness and alters sharp-wave ripple properties.

(A) Firing rate (Hz) and burstiness (P_{burst}; logit-transformed) of Control (black) and MCU-KO (gold) neurons during spontaneous baseline activity. Recordings from MEA in hippocampal cultures (top left schematic). Marginal histograms display the distribution for each axis, with dashed lines indicating group means. Whereas firing rate does not differ significantly between groups ($p = 0.75$), burst probability is significantly higher in MCU-KO neurons ($p = 0.002$).

(B) Same as (A) for tetrode recordings in CA1 of freely-moving mice (top left schematic). Whereas firing rate does not differ significantly between groups ($p = 0.45$), burst probability is significantly higher in MCU-KO neurons ($p = 0.002$).

(C) Wide-band LFP traces of SWR events aligned to the ripple peak ($t = 0$). Data represent mean ± SEM across events. Asterisk denotes a significant increase in peak ripple amplitude ($p = 0.045$), calculated from the bandpass-filtered signal (80-250 Hz).

(D) Instantaneous frequency of SWRs, calculated via Hilbert transform of the bandpass-filtered signal. MCU-KO exhibit a significantly faster oscillation frequency compared to Controls ($p = 0.001$).

(E) Probability distribution of SWR durations. MCU-KO events are significantly shorter than Controls ($p = 0.026$).

(F) Modulation of per-unit firing rates during SWRs (z-scored to baseline). Traces represent mean ± SEM across units.

(G) Probability distribution of spike timing during SWRs, defined as the center of mass relative to the SWR peak (dotted vertical line). Spiking in MCU-KO neurons is significantly advanced (shifted to negative time lags) compared to Controls ($p = 0.027$; Table S1).

(H) Marginal effect of baseline burstiness on the spike center of mass during SWRs, calculated while holding baseline firing rate constant at its mean. Shaded regions indicate the 95% confidence intervals for the predicted means.

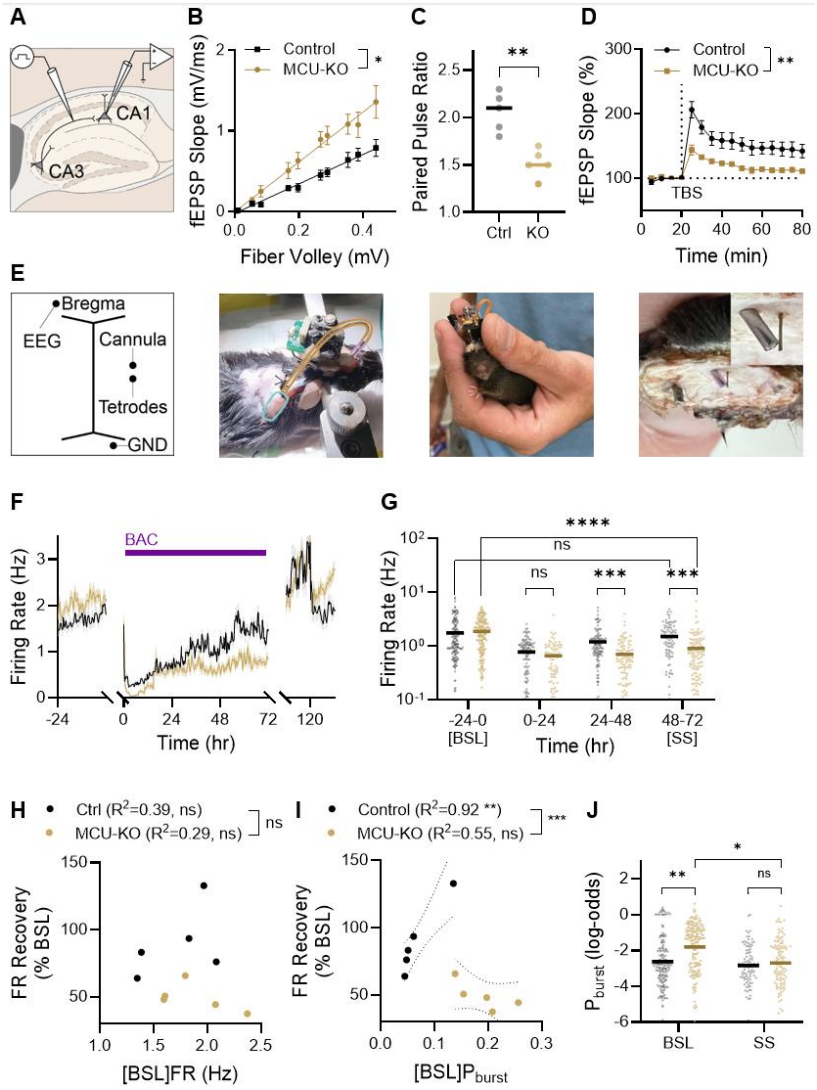


Figure 3. MCU-KO impairs synaptic and homeostatic plasticity.

(A) Schematic illustration of the experimental setup for recording field excitatory postsynaptic potentials (fEPSPs) in the CA1 region of acute hippocampal slices. A stimulating electrode is placed in the Schaffer collateral pathway originating from CA3, and a recording electrode is positioned in the stratum radiatum of CA1.

(B) Input-output (I/O) curves showing the fEPSP slope (mV/ms) as a function of fiber volley amplitude (mV) in Control (n = 5) and MCU-KO (n = 6) slices. MCU-KO slices exhibit a significantly enhanced fEPSP slope for a given FV amplitude (p = 0.01), suggesting increased presynaptic release.

(C) Paired-pulse ratio calculated as the ratio of the second fEPSP slope to the first with an inter-stimulus interval of 50 ms. Individual data points represent slices. MCU-KO slices show a significantly lower PPR compared to Control slices (p = 0.008), indicative of an increased probability of neurotransmitter release.

(D) Long-term potentiation (LTP) at CA3-CA1 synapses. The fEPSP slope is plotted as a percentage of baseline over time (horizontal dashed line). Theta-burst stimulation (TBS) was delivered at t=20 min (vertical dashed line). MCU-KO slices exhibit significantly impaired LTP compared to Control slices (p = 0.006). TBS consisted of 5 bursts at 100 Hz for 250 ms, with a 30-second inter-burst interval.

(E) Methodology for long-term continuous drug delivery to CA1 *in vivo*. From left to right: Schematic diagram illustrating the planned stereotaxic coordinates for the implants relative to Bregma. Locations for EEG screw, guide cannula, tetrodes, and ground (GND) screws are indicated.

Photograph of a mouse head-fixed in a stereotaxic apparatus during the surgical implantation procedure. The outline highlights the osmotic pump (teal), catheter (orange), and guide cannula (pink). Photograph of a mouse post-surgery, held in hand, showing the final assembled head-mounted implant. Post-mortem visualization of implant placement. Photograph of the exposed mouse cranium showing the guide cannula and tetrodes at their final position. The inset provides an enlarged view of the guide cannula and tetrodes.

(F) Firing rate dynamics of Control (N = 5; black) and MCU-KO (N = 5; gold) mice during baclofen perturbation (purple bar). Data represent mean ± SEM across units. Breaks in the time axis mark the implantation and removal of the osmotic pump.

(G) Firing rates of individual units averaged over 24-hour epochs (experimental days) relative to baclofen onset (t=0). Horizontal bars indicate the mean. While control neurons recover steady-state firing equal to baseline (48-72 hr; $p = 0.16$), MCU-KO neurons remained depressed throughout the perturbation ($p < 0.0001$). Correspondingly, the interaction between genotype and time is symmetric initially ($p = 0.30$) but becomes significant at day two (24-48 hr; $p = 0.0001$) and day three (48-72 hr; $p = 0.0003$) of baclofen treatment. Complete LME model presented in Table SX.

(H) Relationship between baseline firing rate and firing rate recovery, expressed as a percentage of baseline. Each data point represents the aggregate mean across units per mouse. ANCOVA revealed no significant effects.

(I) Relationship between baseline burstiness and firing rate recovery. ANCOVA revealed a significant interaction between genotype and burstiness ($p = 0.001$). Dashed lines mark the 95% confidence interval of a linear regression fit per group (Control: $p = 0.009$; MCU-KO: $p = 0.15$).

(J) Burstiness (P_{burst}; logit-transformed) of individual units at baseline (BSL) and steady-state (SS; after three days of baclofen perturbation). Horizontal bars indicate the mean. In contrast to baseline, MCU-KO neurons show no difference in burstiness compared to Controls ($p = 0.59$).

Panels (B,D) analyzed using two-way RM ANOVA with Sidak's multiple comparisons test. Panel (C) analyzed using the Mann-Whitney test.

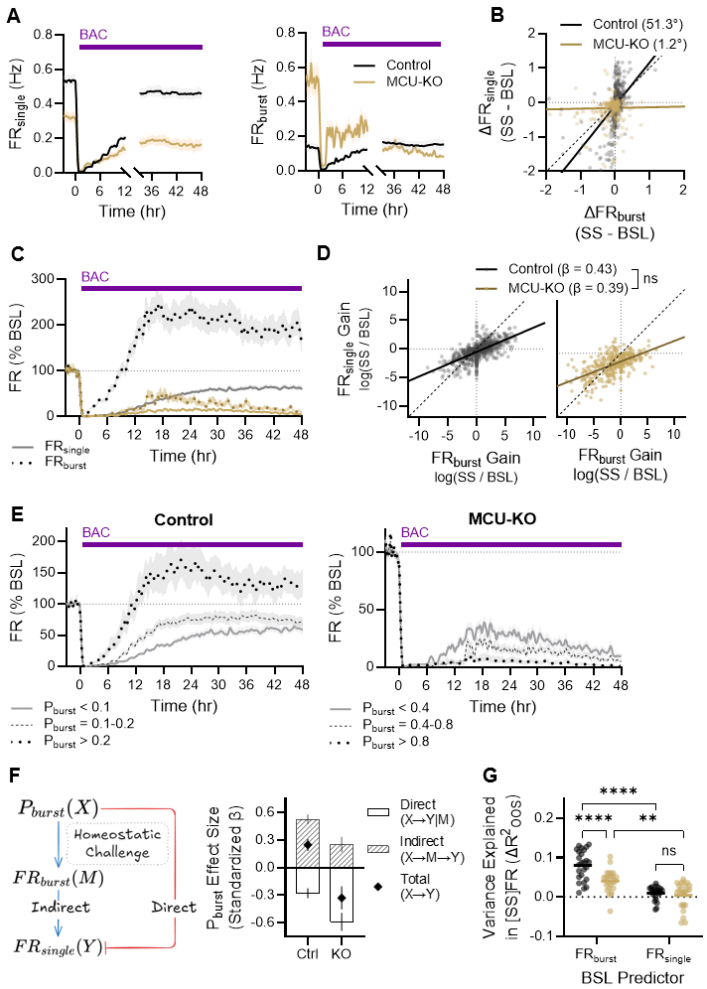


Figure 4. Burstiness is a prognostic marker for homeostasis.

(A) Temporal dynamics of single-spike firing (FR_{single}; left) and burst-spike firing (FR_{burst}; right) during the baclofen perturbation (purple bar).

(B) Absolute differences in single-spike firing (ΔFR_{single}) and burst-spike firing (ΔFR_{burst}) between steady-state (SS) and baseline (BSL). Each point represents a single unit. Dotted crosshairs mark the point of no change. Solid lines depict orthogonal regression fits. The annotated angle quantifies the predominant direction of rate changes relative to the unity line (dashed), which represents a proportional modulation of both components.

(C) Temporal dynamics of single-spike firing (solid) and burst-spike firing (dotted) during the baclofen perturbation. Each neuron's trace was normalized to its pre-perturbation baseline (dotted line). Data presented as geometric mean ± SEM across the population.

(D) State space analysis of spike allocation during firing rate homeostasis for Control (left) and MCU-KO (right) neurons. Single-spike firing gain as a function of burst-spike firing gain, where gain is calculated as the log-fold ratio between steady-state and baseline. Solid lines depict linear regression fits. Burst-spike gain significantly predicts single-spike gain ($\beta = 0.43$; $p < 0.0001$) with no significant interaction by genotype ($\beta = -0.04$; $p = 0.22$). A significant negative main effect of genotype ($\beta = -0.97$; $p = 0.001$) shifts the trajectory of MCU-KO neurons vertically downward. Complete LME model presented in Table SX.

(E) Firing rate dynamics during the baclofen perturbation for neurons stratified by baseline burstiness. Control units (left) were categorized into low (< 0.1; n = 389; dotted), intermediate (0.1-0.2; n = 234; dashed), and high (> 0.2; n = 78; solid) burst probability. MCU-KO units (right) were categorized into low (< 0.4; n = 211), intermediate (0.4-0.8; n = 126), and high (> 0.8; n = 42) burst probability. Data are expressed as a percentage of baseline (dotted line) and presented as geometric mean ± SEM.

(F) Mediation analysis of the relationship between baseline burstiness and steady-state firing. A series of three LME models were fitted separately for Control and MCU-KO neurons to disentangle the direct effect of baseline burstiness (X; [BSL]Pburst) on single-spike firing (Y; [SS]FRsingle) from its indirect effect mediated by burst-spike firing (M; [SS]FRburst). All models included baseline firing rate as a covariate. Stacked bars represent standardized coefficients (β) for the direct (blank) and indirect (stripes) pathways. Black diamonds indicate the net total effect. Vertical lines mark the SEM of each effect. In control neurons, a minimal direct effect ($\beta = -0.278 \pm 0.051$, $p < 0.0001$) is outweighed by a positive indirect effect ($\beta = 0.528 \pm 0.048$, $p < 0.0001$), resulting in a net positive total effect ($\beta = 0.247 \pm 0.063$, $p < 0.0001$), where higher baseline burstiness predicts robust recovery of tonic firing. Conversely, MCU-KO neurons exhibit a severe negative direct effect ($\beta = -0.591 \pm 0.095$, $p < 0.0001$) that negated the indirect coupling ($\beta = 0.252 \pm 0.08$, $p = 0.001$), leading to a net negative total effect and a collapse in tonic firing ($\beta = -0.33 \pm 0.123$, $p < 0.0001$). Complete LME models presented in Table SX.

(G) Feature ablation analysis, utilizing k-fold cross-validation, performed on an LME model predicting steady-state firing rate from baseline firing components ([SS]FR ~ [BSL]FRburst + [BSL]FRsingle + (1|Name)). The y-axis quantifies the unique variance explained by each predictor, calculated as the absolute reduction in the out-of-sample coefficient of determination (ΔR2OOS) when that term is removed from the full model (Control R2OOS = 0.29; MCU-KO R2OOS = 0.26). Dots represent individual cross-validation folds. Horizontal bars indicate the mean. FRburst contributes significantly more unique information than FRsingle across both groups, and significantly more in control than MCU-KO. Two-way RM ANOVA followed by Fisher's LSD post-hoc tests.

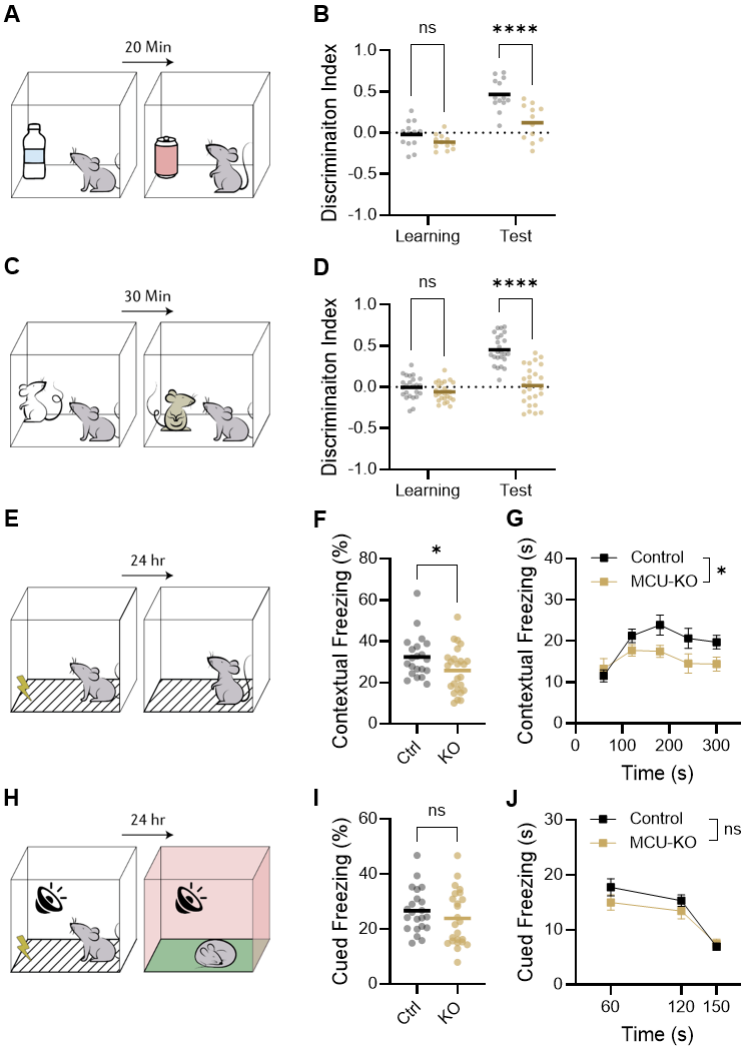


Figure 5. MCU-KO impairs hippocampal-dependent memory.

(A) Schematic of the Novel Object Recognition (NOR) task. During the learning phase, mice explore an arena containing two identical objects (one depicted for simplicity). During the test phase (after a 20-minute inter-trial interval), one object is replaced with a novel object, and exploration times of both objects are recorded.

(B) Discrimination Index for the NOR task. Individual data points represent mice (Control: n = 13; MCU-KO: n = 12), and horizontal bars indicate the group mean. Whereas both groups spent equal time exploring the two objects during learning ($p = 0.45$), control mice showed a preference for the novel object during the test phase that was completely absent in MCU-KO mice ($p = 0.004$).

(C) Schematic of the Social Recognition task. During the learning phase, mice interact with a juvenile conspecific mouse. During the test phase (after a 30-minute inter-trial interval), mice are re-exposed to the familiar juvenile along with a novel juvenile conspecific, and interaction times are recorded.

(D) Discrimination Index for the Social Recognition task. Individual data points represent mice (Control: n = 18; MCU-KO: n = 17), and horizontal bars indicate the group mean. During the learning phase, both groups spent equal time exploring the two mice ($p = 0.45$). During the recognition phase, control mice showed a preference for the novel mouse that was completely absent in MCU-KO mice ($p = 0.004$).

(E) Schematic of the Contextual Fear Conditioning (CFC) task. Mice placed in a conditioning chamber receive 3 electrical shocks (represented by a lightning bolt). During the test phase (after 24 hours), mice return to the same conditioning chamber, and freezing behavior is measured.

(F) Percentage of time spent freezing during the contextual fear memory test phase. MCU-KO mice (n = 25) display significantly lower contextual freezing compared to Control mice (n = 21; $p = 0.043$).

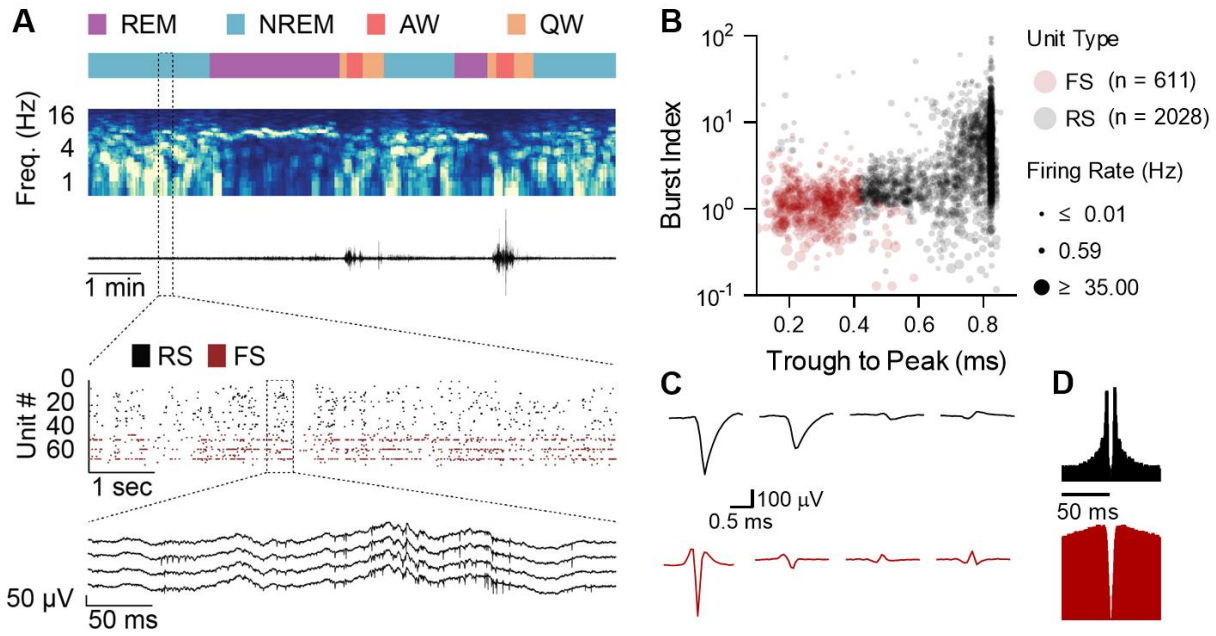
(G) Temporal dynamics of freezing behavior during the CFC test phase, presented as mean ± SEM. MCU-KO mice exhibited reduced freezing levels compared to control mice ($p = 0.043$).

(H) Schematic of the Auditory Cued Fear Conditioning task. Mice placed in a conditioning chamber where an auditory cue (represented by a speaker icon) is paired with electrical shocks. During the test phase (after 24 hours), mice are placed in a novel context (altered visual and olfactory cues), and freezing behavior is measured during the presentation of the auditory cue alone.

(I) Percentage of time spent freezing during the cued fear memory test in the novel context. No significant difference in cued freezing was observed between Control ($n = 22$) and MCU-KO mice ($n = 24$; $p = 0.23$).

(J) Temporal dynamics of freezing behavior during the cued fear memory test phase. No significant difference was found between groups ($p = 0.31$)

Panels (B,D,G,J) analyzed using a two-way RM ANOVA followed by Sidak's multiple comparisons test. Panel (F) analyzed using an unpaired t test. Panel (I) analyzed using the Mann-Whitney test.



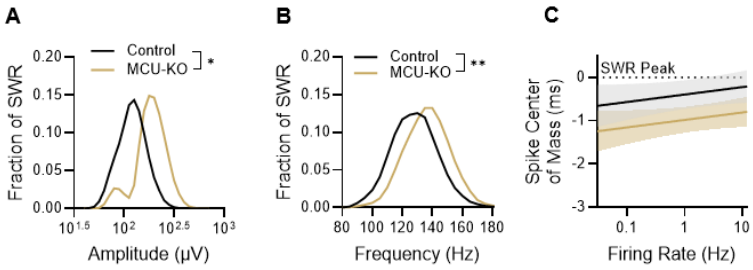
Supp. Figure 1. Extracellular recordings from CA1 in behaving mice.

(A) Representative baseline data from a control mouse. From top to bottom: hypnogram; spectrogram; EMG recordings; raster plot of RS (black) and FS (red) units; raw traces from a single tetrode. Dashed lines indicate an expanded view of the section above.

(B) Trough-to-Peak time (ms) and Burst Index (a.u.) of RS and FS units from MCU-KO and Control mice. Individual data points represent neurons with point size proportional to their firing rate. Assignment of well-isolated clusters to RS and FS was done using a Gaussian Mixture Model (GMM) based on waveform features. FS units were excluded from this study due to low unit count.

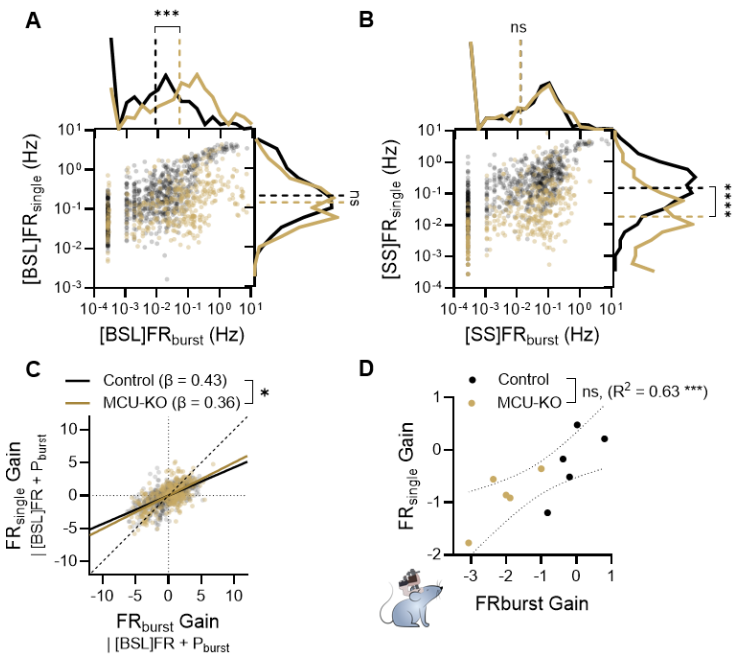
(C) Representative waveform of an RS unit (top) and an FS unit (bottom) across four channels of a tetrode.

(D) Representative autocorrelograms of an RS unit (top) and an FS unit (bottom).



Supp. Figure 2. MCU-KO alters SWR properties.

(A) Marginal effect of baseline firing rate on the spike center of mass during SWRs, calculated while holding baseline burstiness constant at its mean. Shaded regions indicate the 95% confidence intervals for the predicted means. Firing rate did not significantly predict spike timing during SWR ($p = 0.1$; Table SX).



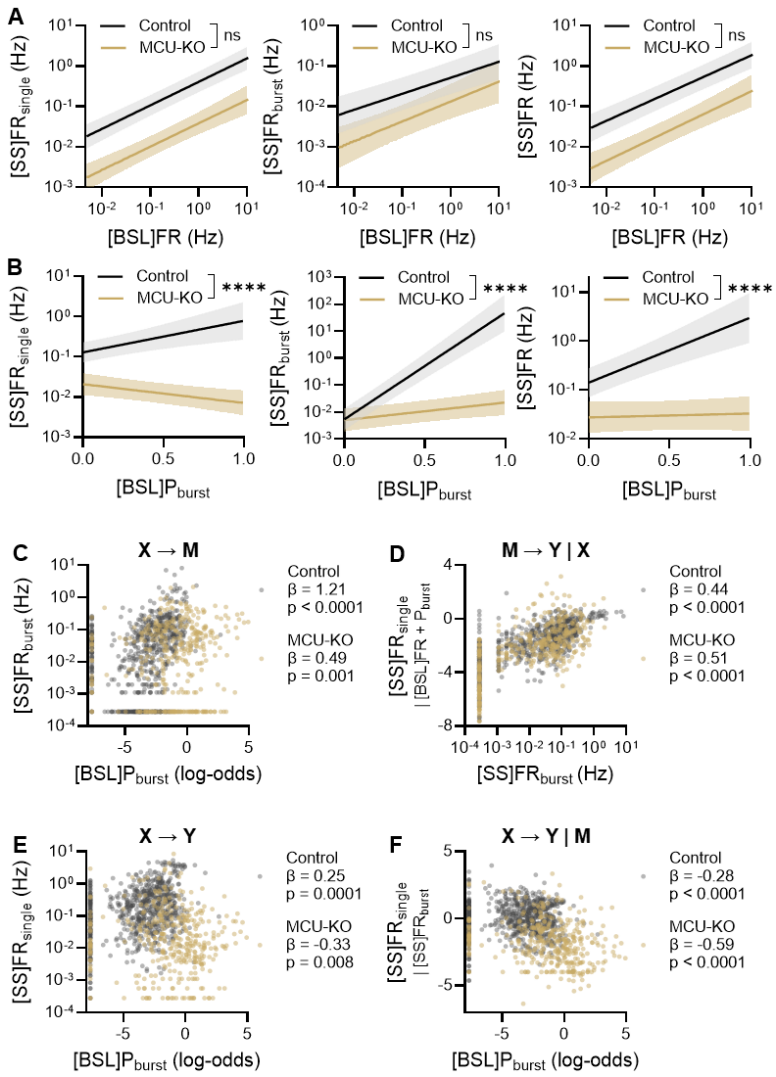
Supp. Figure 3. Homeostatic allocation of spikes

(A) Burst and single spike firing rates in Control (black) and MCU-KO (gold) neurons during baseline (BSL). Marginal histograms display the distribution of firing rates for each axis, with dashed lines indicating the group means.

(B) Same as (A) during steady-state (SS), after two days of baclofen.

(C) Partial regression plot showing the unique variance shared between log-fold changes in single-spike firing (FR_{single} Gain) and burst-spike firing (FR_{burst} Gain). To isolate the specific coupling between these variables, residuals were calculated by regressing out baseline firing rate and burstiness from each component. Solid lines depict linear regression fits. Dotted crosshairs mark the zero-residual center. Burst-spike residuals significantly predicts single-spike residuals ($\beta = 0.43$; $p < 0.0001$). A small but significant interaction by genotype ($\beta = +0.07$; $p = 0.035$) indicates this coupling is stronger in MCU-KO neurons. Because the data represents mean-centered residuals, the regression lines cross exactly at zero with no main effect of genotype ($\beta = 0$; $p = 1$). Complete LME model presented in Table SX.

(D) Spike allocation *in vivo*. Relationship between burst-spike gain and single-spike gain, where gain is calculated as the log-ratio of the average rate during steady-state relative to baseline, and the average rate is the aggregate mean across units per mouse (N = 5 animals per group). ANCOVA revealed a significant main effect of burst gain ($p = 0.034$) but not of genotype ($p = 0.59$) and no interaction ($p = 0.53$). Consequently, a pooled regression model confirmed a conserved positive scaling rule across mice ($p = 0.006$). Dashed lines mark the 95% confidence interval of the linear regression fit.



Supp. Figure 4. Burstiness but not firing rate predicts firing rate recovery

(A) Marginal effects of baseline firing rate on steady-state firing, calculated while holding baseline burstiness constant at its mean. Effects are shown for single-spike firing (left), burst-spike firing (middle), and total firing rate (right). Shaded regions indicate the 95% confidence intervals for the predicted means.

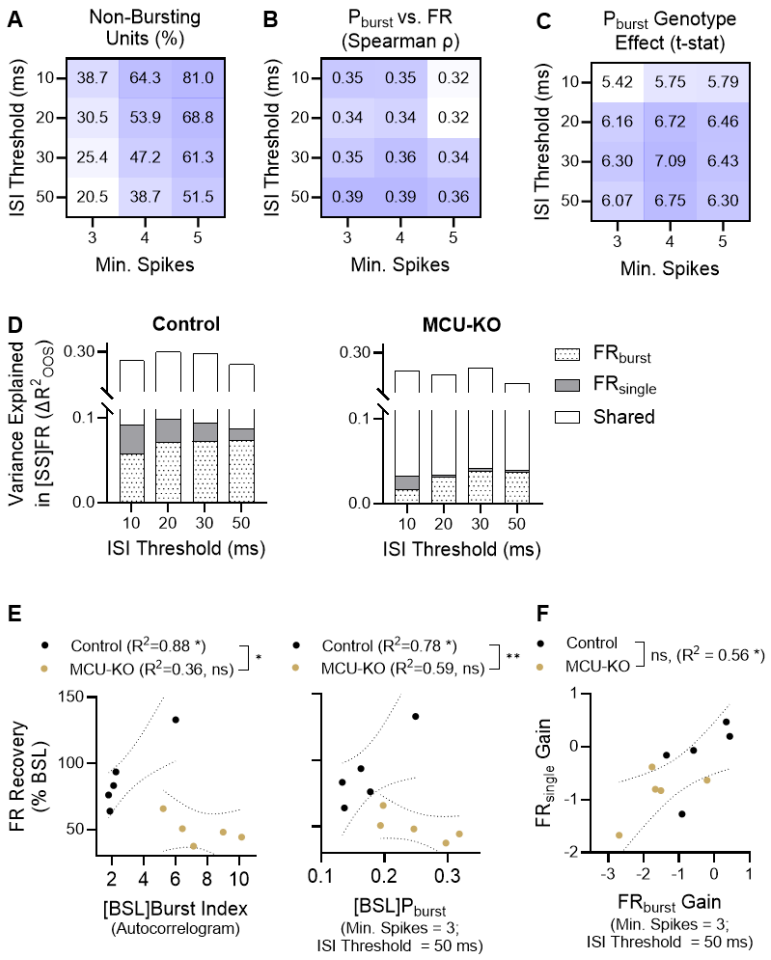
(B) Same as (A) for the marginal effects of baseline burstiness, calculated while holding baseline firing rate constant at its mean.

(C) (X → M): Relationship between baseline burstiness and steady-state burst-spike firing. β derived from Model 2: M ~ X + Covariates.

(D) (M → Y | X): Partial residual plot showing the relationship between steady-state burst-spike firing (M; $[SS]FR_{burst}$) and single-spike firing (Y; $[SS]FR_{single}$), adjusted for baseline firing rate and burstiness (X; $[BSL]P_{burst}$). The Y-axis represents the variance in single-spike firing unexplained by the baseline phenotype. β represents the standardized coefficient of FR_{burst} from Model 1: Y ~ M + X + Covariates.

(E) (X → Y | M): Partial residual plot showing the direct effect of baseline burstiness on steady-state single-spike firing, adjusted for steady-state burst-spike firing. β represents the standardized coefficient of P_{burst} from Model 1.

(F) (X → Y): The total effect (net prognostic relationship) of baseline burstiness on steady-state single-spike firing. β derived from Model 3: Y ~ X + Covariates.



Supp. Figure 5. The impact of burstiness is stable across various detection parameters.

(A) Parameter sweep of burst detection criteria in MEA recordings, including inter-spike interval (ISI) thresholds (y-axis) and minimum number of spikes (x-axis). Labels show the percentage of Control units with zero detected bursts during baseline.

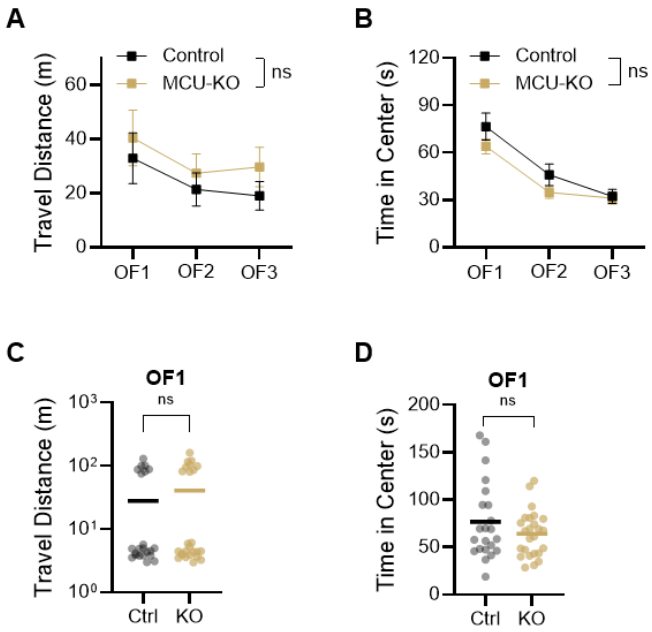
(B) Same as (A) showing the Spearman rank correlation coefficient between baseline firing rate and burstiness in Control units. Firing rate was included as a covariate in all models involving burstiness.

(C) Same as (A) showing the t-statistic for the effect of genotype on burstiness. Positive values indicate significantly higher burstiness in MCU-KO neurons.

(D) Replication of the feature ablation analysis with various ISI thresholds for detecting bursts (minimum spike count fixed at 3). Stacked bars display the unique variance (ΔR^2_{OOS}) in steady-state firing rate explained by baseline burst-spike firing (dotted), single-spike firing (solid gray), and their shared contribution (white). Related to Figure 4G.

(E) Replication of the prognostic relationship between baseline burstiness and firing rate recovery *in vivo* using alternative burst metrics. Left: Relationship using a Burst Index derived from the spike-train autocorrelogram. Right: Relationship using the same criteria for detecting bursts as in the MEA dataset. Related to Figure 3J.

(F) Replication of the homeostatic allocation of spikes in vivo using the same criteria for detecting bursts as in the MEA dataset. Related to Supp. Figure 3E.



Supp. Figure 6. MCU-KO does not affect locomotor activity and anxiety-like behavior.

(A) Total distance traveled (meters) by Control (n = 25) and MCU-KO (n = 22) mice across three consecutive 5-minute Open Field sessions (OF1, OF2, OF3). Both groups traveled significantly less distance across sessions, indicating habituation ($p < 0.0001$). However, there was no significant difference in travel distance between groups ($p = 0.46$) and no significant interaction between group and session ($p = 0.63$).

(B) Total time (seconds) spent in the center zone of the open field, an indicator of anxiety-like behavior. Both groups exhibited a significant decrease in center time across sessions, reflecting habituation ($p < 0.0001$). However, there was no significant difference between groups ($p = 0.21$) and no significant interaction between group and session ($p = 0.27$).

(C) Total distance (meters) traveled by Control and MCU-KO mice during the first Open Field session (OF1). No significant difference in locomotor activity is observed between Control and MCU-KO mice ($p = 0.22$).

(D) Total duration spent in the center zone of the open field (s) by Control and MCU-KO mice during the first Open Field session. Individual data points represent mice. No significant difference in center time is observed between Control and MCU-KO mice ($p = 0.45$).

Panels (A,B) analyzed using two-way RM ANOVA. Panels (C,D) analyzed using the Mann-Whitney test.